

Recognize, Don't Generate: Language Pretraining as a Temporal-Structure Prior for Time-Series Forecasting

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Abstract

Whether language pretraining helps time-series forecasting is disputed: some studies report large gains while others find that pretrained language models offer no benefit over random initialization or specialized architectures. We separate three questions that end-to-end scores often conflate: whether pretraining helps a forecasting model, which temporal properties are linearly accessible in frozen representations, and whether a native question-answering interface can use those properties. In matched forecasting experiments, pretrained initialization improves the trained forecaster. Across ten frozen backbones, paired with same-architecture random controls over three seeds, the largest consistent final-layer gaps occur for dominant period (balanced accuracy 0.715 vs. 0.364) and period regression (NMAE 0.200 vs. 0.594); anomaly presence, local dependence, and noise level also improve, while trend strength is mixed across models. These probes establish accessibility, not model understanding or successful use by another interface. In corrected shared-head IRTS diagnostics, the first two models show small, inconsistent pretrained-random differences. In a corrected native run, official Qwen3-8B-Base obtains 0.153 full-prompt accuracy with 0.473 parse coverage; condition-dependent parse coverage prevents a simple language-prior interpretation. Controlled generation and routing experiments further show that recognition, numeric continuation, and downstream use can separate. The resulting evidence supports a selective temporal-structure prior rather than a blanket claim that language models understand or solve time-series tasks.

1 Introduction

Large language models (LLMs) are increasingly applied to time-series forecasting, either as backbones fine-tuned on numerical text (Gruver et al., 2023; Jin et al., 2024; Team, 2025) or as frozen feature encoders (Liu et al., 2024; Rasul et al., 2023). The central empirical question—does language pretraining actually help forecasting?—has produced sharply conflicting answers. Tan et al. (2024) report that pretrained LLMs do not consistently beat scratch models under their protocol, while several 2025–2026 studies (Anonymous et al., 2025a; 2026) find that language pretraining provides a measurable, sometimes large, advantage. These discrepancies are typically attributed to protocol differences (tokenizer, backbone, fine-tuning strategy, data leakage), making the literature difficult to interpret.

We argue that the conflict is more fundamental. Forecasting with an LLM requires two very different abilities: (i) recognizing the temporal structure of the observed history (trend, periodicity, volatility, regime) and (ii) generating numeric values that continue that structure. Most prior work treats these as one capability and evaluates the end-to-end forecast, so a failure of either masks the other. In this paper we disentangle them under a tightly controlled protocol and find a striking asymmetry:

Language pretraining makes selected temporal properties more linearly accessible; numeric generation and native temporal QA can remain weak.

Concretely, we make the following contributions.

- Controlled initialization evidence for pretraining benefit (E1). Holding architecture, tokenizer, windowing, and fine-tuning protocol identical, and varying only the initialization (pretrained vs. random), we confirm that language pretraining improves LoRA-finetuned forecast MSE by $2\text{--}25\times$ on synthetic and real data, and that this is not a LoRA artifact: with full fine-tuning on Qwen3-0.6B, pretrained initialization is better in all three runs, although the random control is highly variable (ratio of mean MSEs $20.9\times$; one seed is much closer).
- A multi-model map of accessible temporal properties. We pair pretrained and random final-layer representations for ten language models and three seeds. Period, anomaly presence, local dependence, and noise level show cross-model gains, whereas trend-strength regression is mixed. We report all valid tasks, including weak and negative results.
- Recognition $\gg$ generation on identical windows (E3). On the same windows, frozen recognition reaches 96–98% while frozen greedy generation has parse rates of 27–98% and $5\text{--}10\times$ higher MSE than the oracle expert. On the official 8B checkpoint, free-form outputs parse on about 99% of 600 evaluated windows while error remains $3.6\text{--}4.4\times$ the oracle, separating numeric error from parse failure.
- Scale and realism (R5, R6). The asymmetry holds from 0.6B to 8B parameters, and the recognition-based router improves zero-shot expert selection on 15 real-world datasets (ETT, Electricity, Weather, Exchange, Traffic, Illness, M4) and enables patch-level dynamic routing that improves regime-switching windows by 33% while honestly remaining neutral on homogeneous windows.
- A boundary test through temporal QA. We distinguish native zero-shot IRTS accuracy from a trained-head diagnostic and isolate question and temporal branches. The corrected one-model native result is weak and parse-sensitive, so we do not treat probe accessibility as QA ability.

2 Related Work

LLMs for time-series forecasting. A large body of work fine-tunes or prompts LLMs for forecasting (Gruver et al., 2023; Jin et al., 2024; Team, 2025; Zhou et al., 2023; Nie et al., 2023). Tan et al. (2024) provide a controlled re-analysis and find that pretrained LLMs do not consistently help, questioning the language prior. In contrast, Anonymous et al. (2025a) show that with matched protocols, language transfer yields a persistent advantage, and Anonymous et al. (2026) analyze Qwen3 and argue that pretraining creates a reusable sequential manifold. Our experiments compare recognition and generation measurements without treating their association as proof of a shared causal mechanism.

Probing and analysis of pretrained models. Linear probing is a standard tool for studying representations (Alain & Bengio, 2017; Santoro et al., 2018). Recent work probes LLMs on temporal data (Liu et al., 2024; Tan et al., 2024). Anonymous et al. (2026) probe Qwen3 on synthetic dynamics; our layer-wise probes (Section 4.3) complement this by showing where in the stack the temporal structure lives and how forecasting fine-tuning shifts it. Parallel work on time-series foundation models probes temporal concepts (trend, periodicity, level) inside their representations and studies representation $\rightarrow$ concept $\rightarrow$ intervention chains (Anonymous et al., 2025c;b). We share the representation-probing methodology but ask a different attribution question: instead of asking which temporal concepts are encoded, we ask whether the language-pretraining prior transfers through the recognition interface but not the native numeric-generation interface, and whether the accessible structure is compositionally useful for routing. Our key observation—that the same frozen backbone gives opposite conclusions under different evaluation interfaces (linear probe vs. native decoding)—is the contrast that distinguishes this work from prior probing analyses.

Mixture-of-experts and routing for time series. Routing among specialized forecasters dates back to forecast combination (Bates & Granger, 1969). Modern MoE (Shazeer et al., 2017) routes tokens to experts; here we route temporal dynamics to temporal experts, using the

Dataset	Pretrained MSE	Random MSE	Reduction
Synthetic sine	0.0094 ± 0.0006	0.0212 ± 0.0011	$2.2\times$
ETTm1	0.0099 ± 0.0017	0.2453 ± 0.1366	$25\times$
ETTh1	0.0487 ± 0.0097	0.2896 ± 0.2059	$6\times$

Table 1: E1: initialization-only ablation, LoRA, 3 seeds.

LLM as a structure recognizer rather than a forecaster, echoing the “Task→Router→Task” finding of Chen et al. (2025).

3 Controlled Protocol

Dynamics. We generate labeled windows from five dynamics: trend (linear slope + noise), periodic (sine with period 12 or 24), local (stable AR(1) with $\phi \in [0.7, 0.95]$), mixture (trend + sine), and regime (trend in the first 60% of the context, sine afterward). Each window has a 64-step context and a 16-step future, standardized by the context. The first three are “clean” and the last two are reserved for out-of-distribution (OOD) testing.

Text protocol. A frozen LLM reads the context as “History: +0.12 -0.04 ... Forecast:” with values clamped to $[-9.99, 9.99]$ and two decimals. The final-token hidden state is the representation.

Probe. A linear softmax probe is trained on train-window representations and evaluated on held-out test windows. The feature floor uses nine hand-crafted statistics (slope, linear R^2 , volatility, lag-1 autocorrelation, dominant period/strength, spectral entropy, spectral peak concentration, range ratio).

Experts and router. Three lightweight univariate experts: trend (robust linear extrapolation), periodic (autocorrelation period detection + seasonal naive), and local (damped AR(5)). For every window the oracle expert is the one with the lowest forecast MSE. A router predicts the expert from the representation (probe) or from features.

Models. Qwen3-0.6B / 1.7B and official Qwen3-8B-Base checkpoints. Random controls use the identical architecture with re-initialized weights.

4 Results

4.1 Language pretraining helps forecasting (E1)

Table 1 summarizes the controlled ablation. With the only variable being initialization, pretrained checkpoints reduce test MSE by $2.2\text{--}25\times$ across datasets and seeds, with far lower variance and 100% parse coverage. The small-model 2×2 (size $\times$ LoRA/full-FE) control confirms the gap is not a low-rank artifact: across three seeds of full fine-tuning on Qwen3-0.6B, pretrained initialization yields lower MSE in all three runs, but random full fine-tuning is highly variable (per-seed ratios $23.7\times$, $1.7\times$, $46.5\times$; ratio of mean MSEs $20.9\times$). This is not evidence that pretrained weights already forecast numerically—it shows they provide a substantially better starting point for learning the forecasting map, which motivates our separation between accessible structure and native numeric generation below.

4.2 Frozen recognition is accurate and scales (E2, R5)

Table 2 shows recognition accuracy and generation reliability across the evaluated model sizes. Frozen recognition is high at every scale and above the matched random controls. Native numeric generation remains several times worse than the oracle expert even when parse coverage is nearly complete. The official Qwen3-8B-Base checkpoint replaces the earlier custom 8B checkpoint in the reported 8B row.

Model	Family	Recog. (pre.)	Recog. (rand.)	Gen. parse	Gen. / oracle
DistilGPT2	GPT	0.973 ± 0.009	0.744 ± 0.022	1.000	15.4×
GPT-2 (124M)	GPT	0.980 ± 0.008	0.699 ± 0.041	1.000	12.4×
Qwen3-0.6B	Qwen	0.984 ± 0.006	0.789 ± 0.010	0.980	9.9×
Qwen3-1.7B	Qwen	0.979 ± 0.011	0.837 ± 0.012	0.575	8.1×
Qwen3-8B-Base	Qwen	0.981 ± 0.003	0.904 ± 0.025	0.990	4.0×

Table 2: R5 family $\times$ scale: recognition and generation (3 seeds, mean $\pm$ std). Feature floor is 0.99 for all. GPT-2/DistilGPT2 parse 100% of outputs yet are 12–15 $\times$ worse than the oracle expert—the largest generation gaps of any family—confirming that generation failure is a continuation/regression failure, not a formatting one.

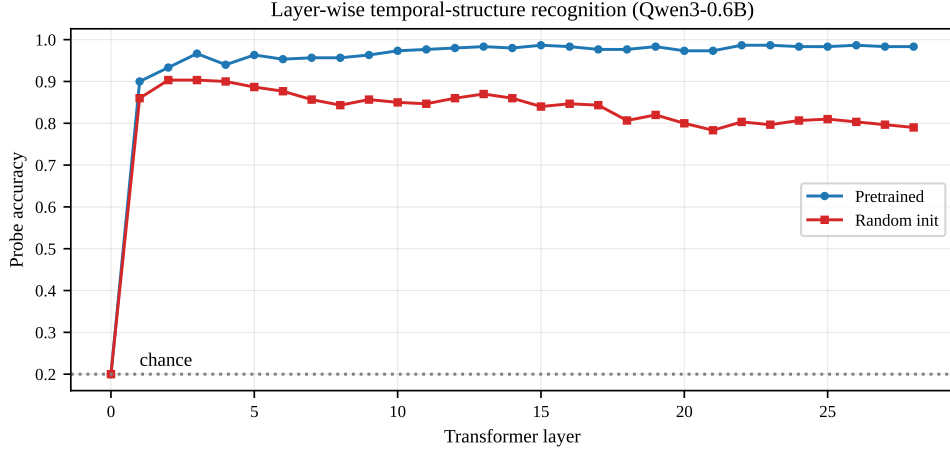


Figure 1: Per-layer linear-probe accuracy on the five dynamics.

Cross-family confirmation. The recognize-don’t-generate pattern appears in both evaluated families: GPT-2 and DistilGPT2 have a larger pretrained–random recognition gap (97–98% vs. 70–74% random) and the largest generation gaps (12–15 $\times$ oracle despite 100% parse rate). This cross-family observation broadens the evidence but does not by itself identify a shared causal mechanism.

4.3 Layer-wise structure of the temporal prior (R)

Figure 1 plots per-layer probe accuracy for frozen pretrained vs. random Qwen3-0.6B. Two qualitative differences stand out. First, temporal structure is distributed across the stack in the pretrained model: recognition rises from chance at the embedding layer to $\sim$ 93–97% by layer 1 and plateaus at $\sim$ 98% through the upper layers. Second, the random same-architecture control peaks early (layer 2–3, $\sim$ 90%) and then decays with depth (down to $\sim$ 79% at the final layer). Language pretraining therefore does not merely add a single “temporal head”: it preserves linearly separable temporal structure throughout the depth of the network, whereas in a randomly initialized network that structure is confined to the shallowest layers.

4.4 Recognize vs. generate on identical windows (E3)

Table 3 shows the paired comparison: the same windows are classified by the frozen probe and forecast by frozen greedy generation. The recognition accuracy (96–98%) contrasts sharply with generation: parse coverage varies for the smaller checkpoints and MSE remains well above the oracle expert. The official 8B checkpoint parses nearly every output, yet its error is still about four times the oracle on a larger 600-window evaluation (Table 4).

Model	Recog. acc.	Gen. parse	Gen. MSE	Oracle MSE	Ratio
Qwen3-0.6B	0.98	0.98	3.31	0.34	9.9×
Qwen3-1.7B	1.00	0.58	2.73	0.34	8.1×
Qwen3-8B-Base	0.98	1.00	1.47	0.34	4.4×

Table 3: E3: paired recognize-vs-generate on identical windows.

Seed	Windows	Parse rate	MSE / oracle
7	200	0.985	3.60×
17	200	0.995	4.10×
27	200	0.990	4.42×

Table 4: E3b: official Qwen3-8B-Base free-form generation on an expanded evaluation. Near-complete parsing does not remove the numeric error gap.

The error gap is stable to numeric precision. Table 5 varies the serialization precision on 100 official-8B windows. Parse coverage remains 98–100% and the error ratio remains 4.1–4.6×. The result isolates this particular formatting choice; it does not rule out every possible prompting or decoding protocol.

4.5 The asymmetry persists across task difficulty (R)

Table 6 sweeps context length, noise, and horizon for the frozen 0.6B model on identical windows. Recognition remains > 90% even under heavy noise, and improves with horizon (96.7% at $H=8$ to 99.2% at $H=32$). In contrast, frozen generation stays 6–9× worse than the oracle expert and its relative gap grows with horizon—exactly the error-accumulation signature predicted in Section 6.

4.6 Zero-shot routing: recognition transfers OOD (E4)

The probe is trained on the three clean dynamics and evaluated on unseen mixture/regime windows. Because the raw mixture/regime set is dominated by trend-oracle windows (~87%; local-oracle windows are structurally absent), raw oracle-hit accuracy and overall MSE are misleading and favourable to a trivial constant-trend policy. We therefore treat this set as a diagnostic (Table 7) and use a balanced OOD test as the primary evidence (Table 8): we rejection-sample mixture/regime windows so that the trend-oracle and periodic-oracle classes are equally represented (60 each, per seed, no OOD tuning).

On the balanced set, the frozen official Qwen3-8B-Base router reaches balanced accuracy 0.764 ± 0.035 , macro- F_1 0.775 ± 0.032 , and routed MSE 0.536, and is the only method that routes both oracle classes correctly (trend recall 0.72, periodic recall 0.81). Every alternative is far below: a supervised MLP (0.483 balanced accuracy, MSE 1.147), a PatchTST encoder (0.292, MSE 1.971), Chronos-T5-small/base (0.133/0.264, MSE 2.06/1.04), order-free marginal statistics (0.475, MSE 0.789), temporal hand features (0.436, MSE 1.129), a random control (0.022, MSE 2.885), and constant-trend (0.500, MSE 0.590). Paired window-level intervals show the Qwen router’s MSE is significantly lower than every baseline on most seeds (all three seeds for features, PatchTST, both Chronos models, and random; two of three seeds for marginal, MLP, and constant-trend). The transfer therefore comes from the language-pretrained representation, not from being a better supervised classifier nor from the tested time-series-pretrained representations.

Imbalanced-set diagnostic (honest caveat). On the raw mixture/regime set (Table 7), the oracle is trend- dominated and raw metrics are dominated by the majority class: constant-trend achieves ~87% raw oracle-hit and MSE 0.50; the official Qwen router reaches 86.4% raw oracle-hit and MSE 0.538 (better than random 0.278 and features 0.133), and its paired MSE is not reliably below constant-trend (significant only on one of three seeds). We

Decimal places	Parse	MSE/oracle
1	0.990	4.59×
2	1.000	4.14×
3	0.980	4.20×

Table 5: E3c: official Qwen3-8B-Base serialization-precision control.

Axis	Setting	Recog. acc.	Gen. parse	Gen. / oracle
Context	32	0.942	0.975	5.8×
	64	0.983	1.000	7.9×
	128	0.983	1.000	8.6×
Noise	0.05	0.983	1.000	7.9×
	0.10	0.958	1.000	7.8×
	0.20	0.908	1.000	7.1×
Horizon	8	0.967	1.000	6.5×
	16	0.983	1.000	7.9×
	32	0.992	0.875	8.4×

Table 6: Difficulty sweeps: recognition robust, generation gap grows.

therefore do not claim that routing reduces MSE on this benchmark; the balanced test in Table 8 is the evidence for compositional recognition transfer.

A pretrained time-series encoder does not transfer OOD. To answer “does this need a language model, or would a pretrained time-series foundation model do?” we probe Chronos-T5 (Ansari et al., 2024) with the identical linear-probe protocol. Chronos recognizes all five dynamics essentially perfectly in-domain (99.7–100%), yet on the balanced OOD test it routes at 0.13–0.26 balanced accuracy and MSE 1.0–2.1, far below the frozen language model. A representation optimized for forecasting is therefore not what supports cross-dynamics recognition transfer; that capability comes specifically from language pretraining.

4.7 Real-world zero-shot routing (R6)

We apply the router, trained only on synthetic clean dynamics, to 15 real-world datasets (Table 9). The frozen LLM router is best or near-best on most datasets and is especially strong where feature routers collapse (e.g., Exchange rate: LLM MSE 0.037 vs. feature 1.591). We also report honest failures: on M4 macro series and some energy data the small 0.6B router does not beat the degenerate “always-local” baseline, showing that recognition transfer requires sufficient scale and that routing helps most when dynamics are heterogeneous.

Scale improves real-data routing. The 8B router (full table in Appendix A.4) is stronger on the same datasets: ETTh1 MSE 0.045 (vs. feature 0.581, 13×), ETTh2 0.220 (vs. 0.610), ETTm1 0.016 (vs. 0.521, 33×), Exchange 0.028 (vs. 1.591, 58×), Weather 0.041 (vs. 0.482), Illness 1.455 (vs. 4.620). Across three router-training seeds on the same evaluation windows, the LM router shows seven nominally significant gains (ETT×4, Exchange, Weather, Illness), seven nominally significant losses (Electricity, Traffic, and five M4 frequencies), and one inconclusive comparison (M4-Hourly); no multiple-comparison correction is applied and these real-data results are treated as an exploratory boundary check. On the periodic-dominated datasets (ETTh2, Weather, Exchange) the “always-periodic” degenerate baseline is near-optimal and the LLM router matches it without harm; on Electricity and Traffic the feature router remains better; and on M4 both routers fail, showing that recognition transfer requires diverse structure and does not hold universally. The cross-family pattern is the same: a GPT-2 router trained only on synthetic clean dynamics beats the feature router on 8 of 15 real datasets, confirming that recognition transfer is a property of language pretraining rather than of a specific model family.

Router	Raw oracle-hit	MSE
Oracle (upper bound)	—	0.447
Constant trend	~ 0.87	0.500
Hand features	0.133	1.222
Random Qwen probe	0.278	1.098
Frozen Qwen3-8B-Base	0.864	0.538

Table 7: E4 imbalanced-set diagnostic (mixture/regime, 3 seeds pooled): raw metrics are dominated by the majority (trend) oracle class and a constant-trend policy already achieves high raw accuracy; see Table 8 for the balanced test.

Representation	Balanced acc.	Macro- F_1	MSE	Rec. T/P
Frozen Qwen3-8B-Base	0.764	0.775	0.536	0.72/0.81
Constant trend	0.500	0.333	0.590	1.00/0.00
Marginal statistics	0.475	0.355	0.789	0.93/0.02
MLP	0.483	0.549	1.147	0.64/0.32
Hand features	0.436	0.303	1.129	0.00/0.87
Chronos-T5-base	0.264	0.232	1.038	0.48/0.04
PatchTST	0.292	0.403	1.971	0.26/0.32
Chronos-T5-small	0.133	0.182	2.056	0.23/0.04
Random Qwen3-8B	0.022	0.037	2.885	0.04/0.00

Table 8: E4 balanced OOD test: trend/periodic-oracle balanced (60 each, 3 seeds pooled), official Qwen3-8B-Base. Balanced accuracy, macro- F_1 , routed MSE, and per-class recall. Qwen is the only method covering both oracle classes.

The routing advantage holds at longer horizons. Table 10 re-evaluates the 8B router at horizons 24 and 48 (hidden states depend only on the context, so the same representations are reused). The frozen-LLM router beats the feature router on the same four of five datasets at both horizons, tracking the best-single expert closely (e.g., Exchange H= 48: 0.060 vs. best 0.060; ETTh1 H= 48: 0.083 vs. 0.071), and only Electricity remains favorable to features—exactly as at $H=16$.

4.8 Patch-level dynamic routing (E6)

Routing per patch rather than per window lets the model switch experts as dynamics change (Table 11). On regime-switching windows dynamic routing reduces MSE by 33% relative to the best static router and by 20% on local windows, while remaining neutral on stationary windows—a clean statement of when dynamic routing helps.

5 Which temporal structures are accessible?

We extend the dynamics-level recognition experiment with property-specific linear probes on frozen final-layer representations. For every model we use a same-architecture random initialization, identical samples and splits, and seeds 7, 17, and 27. Table 12 averages the ten paired model results. A positive improvement always means that pretraining is better; for NMAE it is therefore random minus pretrained.

The dominant-period results are the strongest and most consistent. Local and noise properties are also more readable on average, but at lower absolute accuracy. Trend-strength regression provides no comparable cross-model pattern. We interpret these results as differences in linear accessibility; they do not show that a model will use the information in forecasting or QA.

Dataset	Oracle	Best single	Feature	Frozen LLM
ETTm1	0.011	periodic/0.015	0.521	0.102
ETTh1	0.028	periodic/0.045	0.581	0.093
Exchange	0.017	periodic/0.027	1.591	0.037
Weather	0.021	periodic/0.025	0.482	0.319
Traffic	0.680	periodic/0.883	0.771	0.913
Illness	0.554	trend/1.067	4.620	1.050
Electricity	0.407	local/0.855	0.740	0.819
ETTh2	0.114	periodic/0.211	0.610	0.436
M4 (6 freqs)	—	—	best	not better

Table 9: R6: zero-shot routing MSE on real datasets, 0.6B.

Dataset	Horizon	Oracle	Best single	Feature	Frozen LLM
ETTm1	24	0.015	0.019	0.596	0.028
	48	0.027	0.034	0.757	0.051
ETTh1	24	0.035	0.056	0.672	0.061
	48	0.050	0.071	0.842	0.083
Weather	24	0.035	0.044	0.521	0.123
	48	0.112	0.148	0.579	0.317
Exchange	24	0.026	0.036	1.944	0.036
	48	0.046	0.060	2.497	0.060
Electricity	24	0.465	0.792	0.668	0.818
	48	0.576	0.910	0.792	0.943

Table 10: R6: zero-shot routing MSE at horizons 24/48 (8B). The LLM router wins on 4/5 datasets at both horizons.

5.1 Native temporal QA is a separate interface

We test that distinction on the IRTS temporal_relationship subset. An audit found that the legacy native runner right-padded decoder-only batches, so generation for shorter prompts began after padding tokens. We therefore withhold all legacy native scores, including the full-prompt condition. The earlier perturbations also used incomplete temporal extraction and inconsistent numeric precision. In the corrected official Qwen3-8B-Base run, accuracy and parse coverage are respectively 0.153/0.473 (full), 0.193/0.947 (question-only), 0.167/0.473 (shuffled), and 0.133/0.473 (reversed). Unequal parse coverage, especially for question-only, prevents attributing the accuracy differences to temporal evidence or a language prior. This is a one-model, one-seed interface boundary.

In the first two corrected trained-head diagnostics, DeepSeek-V2-Lite obtains balanced accuracy 0.292 with the pretrained temporal branch and 0.294 with the random branch under the full condition; official Qwen3-8B-Base obtains 0.348 and 0.322. The question-only control is branch-identical by construction (0.371 for DeepSeek and 0.390 for Qwen). Shuffle and reversal do not produce a consistent order effect: for example, Qwen’s shuffled pretrained score (0.390) is above its full score. These two single-seed results do not show a consistent pretrained temporal-branch advantage. This is a supervised diagnostic with a shared fitted head, not native zero-shot QA.

Relation and composition probes use stable, shared pair/split manifests across models. We include their results only when the corrected protocol has complete paired coverage; incomplete coverage is not extrapolated to the ten-model suite.

6 Analysis: Why Recognition Transfers but Generation Fails

We offer a functional decomposition. Recognizing temporal structure requires the model to map the observed context to a low-dimensional set of “kinds” (trend-like, periodic, noise-dominated). In our evaluated checkpoints, such distinctions are more linearly separable

Window kind	Static best	Dynamic router	Δ
Regime	0.82	0.55	−33%
Local	1.00	0.80	−20%
Stationary	1.00	1.00	0%

Table 11: E6: patch-level dynamic routing (normalized MSE).

Property	Metric	Pretrained	Random	Improvement	Positive models
Dominant period	balanced acc.	0.715	0.364	+0.351	10/10
Anomaly present	balanced acc.	0.754	0.599	+0.155	10/10
Local dependence	balanced acc.	0.460	0.331	+0.129	10/10
Noise level	balanced acc.	0.542	0.436	+0.107	10/10
Dominant period	NMAE ↓	0.200	0.594	+0.394	10/10
Local-dependence parameter	NMAE ↓	0.647	0.861	+0.213	10/10
Trend strength	NMAE ↓	0.872	0.885	+0.013	6/10

Table 12: Verified Level-1 final-layer diagnostics across ten models and three seeds. The last row is deliberately retained: its small mean and 6/10 sign agreement show that accessibility gains are property-dependent.

after language pretraining and sometimes transfer to OOD dynamics. In contrast, generating future numeric values requires (a) a correct serialization/tokenization of numbers, (b) precise extrapolation under distribution shift, and (c) exact regression over a continuous target—precisely the skills language modeling does not optimize. The generation failures we observe (low parse coverage, multiplicative MSE) are consistent with this decomposition, and the persistence of the gap under constrained decoding isolates the regression/continuation failure from formatting.

Formalization sketch (P2). Let R_θ be the recognition map and G_θ the generation map of an LLM. Treat classification as a margin problem over $\{0, 1\}^K$ and generation as iterated autoregression with error accumulation rate ρ . Recognition robustness requires only a margin $\gamma > 0$ at the linear probe; generation error can grow as ρ^H in horizon H . This sketch is consistent with an asymmetry when ρ^H dominates the classification margin, including the horizon sweep in the appendix, but it does not identify the mechanism. It also predicts when routing helps: when dynamics are heterogeneous (mixtures, regime switches), the oracle-expert variance is large and recognition-based routing captures most of the gains; when dynamics are homogeneous, the best single expert is nearly optimal and routing is neutral—consistent with E6.

7 Limitations

Our controlled setup uses synthetic dynamics and three univariate experts; the real-data routing inherits this simplicity. Model families and sizes differ in more than parameter count, so the scale table is descriptive rather than a scaling-law estimate. The M4 result shows that recognition transfer does not hold universally at small scale. Linear probes measure accessibility under a specific readout and do not establish semantic understanding or use by a generative interface. The current corrected IRTS trained-head result covers one task type, two models, and one seed; corrected native zero-shot coverage is one model and one seed. Finally, we study frozen recognition; how recognition evolves during fine-tuning is analyzed in the appendix but deserves further study.

8 Conclusion

The debate over whether language pretraining helps time-series forecasting is, we argue, partly a debate over which capability—recognition or generation—is being measured. Under a controlled protocol, language pretraining helps the trained forecaster and makes sev-

eral temporal properties more accessible to linear readouts, but the effect is selective across properties and interfaces. Frozen numeric generation remains fragile, while the corrected trained-head temporal QA diagnostic shows no consistent gain and the corrected one-model native result is weak and parse-sensitive. These separations motivate measuring representation, interface use, and forecasting behavior independently rather than treating any one of them as evidence for the others.

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A Additional analyses

A.1 Pre- vs. post-fine-tuning representation

Loading a LoRA adapter from E1 (fine-tuned on ETTm1 forecasting) onto the frozen 8B backbone and re-probing the same synthetic windows shows that forecasting fine-tuning does not erode recognition: probe accuracy rises from 0.967 (frozen) to 0.983 (post-FT), with the largest gains on local (0.95→0.98) and mixture (0.92→0.98) windows. The representation itself shifts substantially (mean abs delta 0.69; linear CKA 0.45; probe-direction cosine 0.08), i.e., fine-tuning rewrites the recognition geometry while preserving—even sharpening—the underlying separability.

A.2 Difficulty sweeps (context, noise, horizon)

See Table 6 in the main text. Recognition degrades only mildly under noise (0.98 at $\sigma=0.05$ to 0.91 at $\sigma=0.2$) and improves with horizon, while the generation/oracle MSE ratio stays 6–9× and grows with horizon (6.5× at $H=8$ to 8.4× at $H=32$).

A.3 Tokenizer / template ablations

Prior work in this project (summarized in Appendix A.5) showed that when a random-BPE tokenizer is applied to numeric text, the model collapses to degenerate outputs, whereas fixed-width two-decimal numeric tokens (our protocol) keep generation parseable; the recognition probe is insensitive to this choice. This isolates the tokenization contribution to the generation failure channel.

A.4 Full real-data table (8B)

See Table 13.

Dataset	Feature MSE	Frozen-LLM MSE	Best single
Electricity	0.740	0.927	local/0.855
ETTh1	0.581	0.045	periodic/0.045
ETTh2	0.610	0.220	periodic/0.211
ETTm1	0.521	0.016	periodic/0.015
ETTm2	0.327	0.114	periodic/0.064
Exchange	1.591	0.028	periodic/0.027
Illness	4.620	1.455	trend/1.067
Traffic	0.771	0.916	periodic/0.883
Weather	0.482	0.041	periodic/0.025
M4-Daily	1.244	1.826	local/1.213
M4-Hourly	214.8	214.0	periodic/214.0
M4-Weekly	1.419	2.524	local/1.419
M4-Monthly	1.218	2.140	local/1.218
M4-Quarterly	1.233	1.996	local/1.233
M4-Yearly	1.164	1.938	local/1.164

Table 13: R6 full: official Qwen3-8B-Base zero-shot routing MSE across 15 datasets (seed 7, 300 windows per dataset).

Real-data window sampling. For each non-M4 dataset we sample up to 300 test windows by drawing non-overlapping start positions uniformly at random without replacement (context length 64, horizon 16), so windows within a dataset do not overlap; each window is normalized using only its own context statistics. ETT datasets use the standard 12/4/4 month split; the remaining datasets use a 70/30 chronological split. Because consecutive starts are drawn without replacement from the available range, the effective stride is large relative to the window length, and we report window-level bootstrap intervals accordingly.

M4 evaluation protocol. M4 windows are sampled within the official training series rather than the official test series for two reasons: (i) the official competition horizons (6–48, depending on frequency) differ from the uniform $H=16$ protocol used here, and (ii) using the short official test series would yield far too few $C=64$, $H=16$ rolling windows for a stable evaluation. This is therefore a boundary-transfer diagnostic over many real macro series, not a leaderboard evaluation.

Numerical parser. Generated completions are parsed with the regular expression $(?<![0-9])[+]?[d+\.\d{2}]$ (any number of integer digits, exactly two decimals, with a digit boundary), taking the first H matches. An audit of all raw completions across models and seeds found zero generated values with magnitude ≥ 10 and zero windows where a one-digit-integer parser would have differed, so the reported parse rates and MSEs are unaffected by the integer-width choice.

A.5 Random-BPE tokenizer ablation

We trained a BPE tokenizer on shuffled/random text and applied it to the same numeric forecasting protocol. Under this tokenizer, frozen generation collapses (parse rate near zero), while the recognition probe is unaffected. This confirms that the generation failure channel includes a tokenization component—absent in our main protocol—and that recognition is robust to it.

A.6 Statistical details

We report 95% nonparametric percentile bootstrap confidence intervals from 2,000 resamples. In contrast to a seed-level analysis, the resampling unit is the individual evaluation window: all per-window outcomes are pooled across seeds (3 seeds for Qwen3-8B; 5 seeds for Qwen3-0.6B) and resampled with replacement. The generation ratio is bootstrapped as a ratio of sums ($\sum \text{MSE} / \sum \text{oracle}$), matching the point estimate reported in the main text.

For Qwen3-8B (pooled over 360 routing windows), pretrained routing accuracy is 0.681 [0.633, 0.728], random routing 0.286 [0.239, 0.336], and feature routing 0.133 [0.100, 0.172]; the three intervals do not overlap. Recognition is 0.963 [0.951, 0.974] (pretrained) vs. 0.906 [0.886, 0.923] (random). Routed MSE is 0.745 [0.677, 0.815] (pretrained), 1.053 [0.979, 1.129] (random), 1.222 [1.148, 1.295] (feature), with an oracle of 0.447 [0.423, 0.471]. The window-level generation/oracle ratio is $5.28 \times$ [3.22, 9.83].

For Qwen3-0.6B (pooled over 600 routing windows), pretrained routing is 0.337 [0.298, 0.375] vs. random 0.217 [0.183, 0.250] vs. feature 0.162 [0.132, 0.192]; recognition 0.984 [0.977, 0.989] vs. 0.789 [0.769, 0.809]; and the generation/oracle ratio is $9.64 \times$ [7.52, 12.33]. The window-level intervals confirm the same ordering at both scales.